

Digital Communications

Passband Data Transmission

Passband Transmission Model

Coherent Phase-Shift Keying

Hybrid Amplitude/Phase Modulation Schemes



Introduction

- **Passband Data Transmission**
 - These topics builds on the material developed in last session on signal-space analysis
 - It discusses the subject of digital data transmission over a band-pass channel that can be linear or nonlinear
 - As with analog communications, this mode of data transmission relies on the use of a sinusoidal carrier wave modulated by the data stream

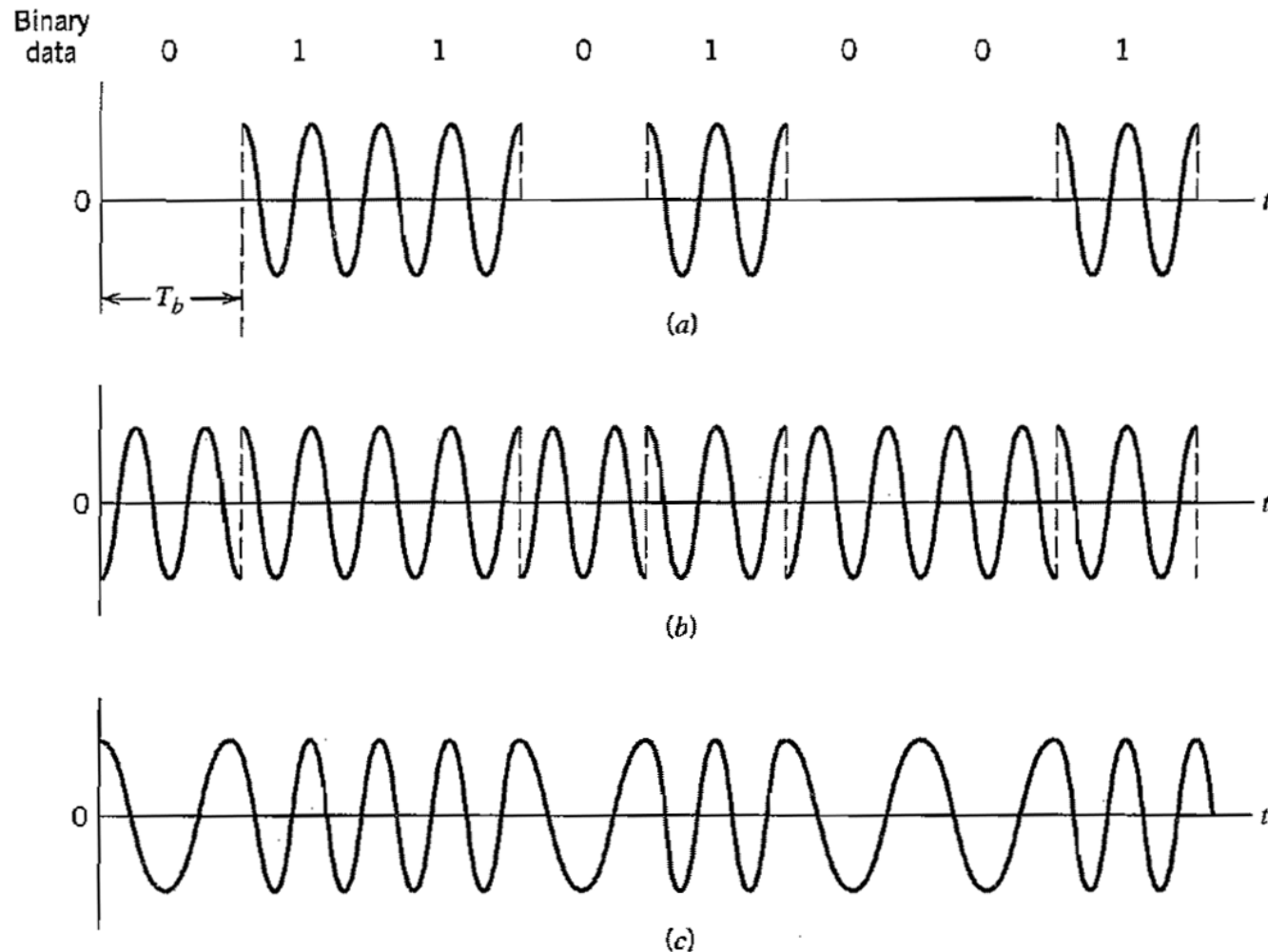


Passband Data Transmission

- In any even the modulation process making the transmission possible involves switching (keying) amplitude, frequency, or phase of a sinusoidal carrier in some fashion in accordance with the incoming data
- Thus there are three basic signaling schemes, and they are known as:
 - Amplitude - Shift Keying (ASK) – (a)
 - Phase - Shift Keying (PSK) – (b)
 - Frequency - Shift Keying (FSK) – (c)



Passband Data Transmission



Passband Data Transmission

- The following points are noteworthy from figure:
 - Although in continuous-wave modulation (CW) it is usually difficult to distinguish between phase-modulated and frequency-modulated signals by merely looking at their waveforms, this is not true for PSK and FSK signals
 - Unlike ASK signals, both PSK and FSK signals have a constant envelope
 - This latter property makes PSK and FSK signals impervious to amplitude nonlinearities, commonly encountered in microwave radio and satellite channels



Passband Data Transmission

- It is for this reason, in practice, we find that PSK and FSK signals are preferred to ASK signals for passband data transmission over nonlinear channels



Passband Data Transmission

- Hierarchy of Digital Modulation Techniques
 - Digital modulation techniques may be classified into coherent and noncoherent, depending on whether the receiver is equipped with a phase-recovery circuit or not
 - The phase-recovery circuit ensures that the oscillator supplying the locally generated carrier wave in the Rx is synchronized (both frequency and phase) to the oscillator supplying the carrier wave used to originally modulate the incoming data stream in the Tx



Passband Data Transmission

- Hierarchy of Digital Modulation Techniques
 - As discussed in previous session, in an M -ary signaling scheme, we may send any one of M possible signals $s_1(t), s_2(t), \dots, s_M(t)$, during each signaling interval of duration T
 - For almost all applications, the number of possible signals $M = 2^n$, where n is an integer
 - The symbol duration $T = nT_b$, where T_b is the bit duration

Passband Data Transmission

- Hierarchy of Digital Modulation Techniques
 - In passband data transmission these signals are generated by changing the amplitude, phase, or frequency of a sinusoidal carrier in M discrete steps
 - Thus, we have M -ary ASK, M -ary PSK, and M -ary FSK digital modulation schemes
 - A special form of this hybrid modulation is M -ary quadrature-amplitude modulation (QAM), which has some attractive properties. M -ary ASK is a special case of M -ary QAM



Passband Data Transmission

- Hierarchy of Digital Modulation Techniques
 - M -ary signaling schemes are preferred over binary signaling schemes for transmitting digital information over band-pass channels when the requirement is to conserve bandwidth at the expense of increased power
 - In practice, we rarely find a communication channel that has the exact bandwidth required for transmitting the output of an information source by means of binary signaling schemes



Passband Data Transmission

- Hierarchy of Digital Modulation Techniques
 - Thus, when the bandwidth of the channel is less than the required value, we may use M -ary signaling schemes for maximum efficiency
 - To illustrate the bandwidth-conservation capability of M -ary signaling schemes consider the transmission of information consisting of a binary sequence with bit duration T_b



Passband Data Transmission

- Hierarchy of Digital Modulation Techniques
 - However, if we take blocks of n bits and use an M -ary PSK scheme with $M = 2^n$ and symbol duration $T = nT_b$
 - The bandwidth required is proportional to $1/nT_b$
 - This shows that the use of M -ary PSK enables a reduction in transmission bandwidth by the factor:
 - $n = \log_2 M$ over binary PSK



Passband Data Transmission

- Hierarchy of Digital Modulation Techniques
 - M -ary PSK and M -ary QAM are examples of linear modulation. However, they differ from each other in one important respect:
 - M -ary PSK signal has a constant envelope, whereas an M -ary QAM signal involves changes in the carrier amplitude
 - Thus, M -ary PSK can be used to transmit digital data over a nonlinear band-pass channel, whereas M -ary QAM requires the use of a linear channel

Passband Data Transmission

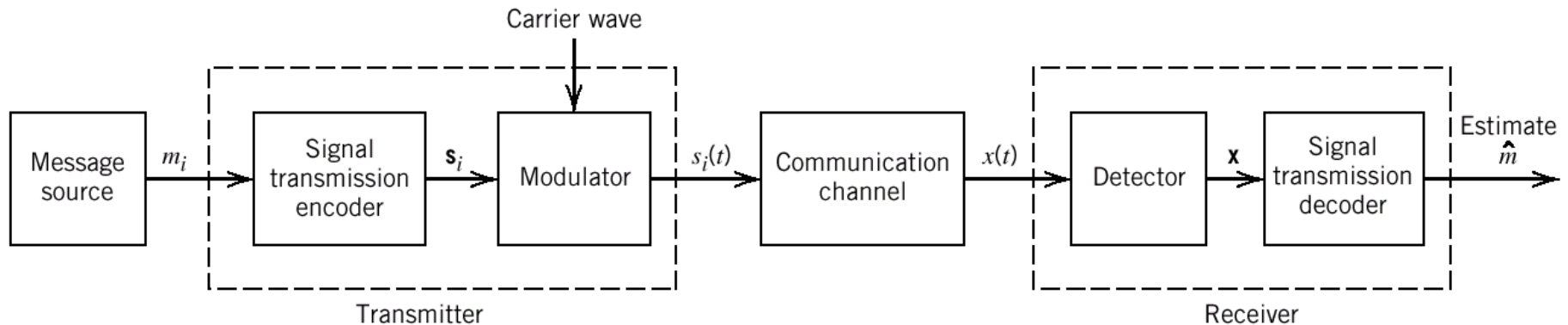
- Hierarchy of Digital Modulation Techniques
 - M -ary PSK, M -ary QAM, and M -ary FSK are commonly used in coherent systems
 - ASK and FSK lend themselves naturally to use in non-coherent systems whenever it is impractical to maintain carrier phase synchronization
 - But in the case of PSK, we cannot have noncoherent PSK because the term noncoherent means doing without carrier phase information



Passband Data Transmission

- **Passband Transmission Model**

- We assume a linear AWGN channel
- Receiver estimates the transmitted symbols, minimizing the effect of channel noise



Passband Data Transmission

- **Single carrier modulations:**
 - Waveforms
 - Basis functions
 - Constellations
 - Error probability
 - Power spectrum and bandwidth
 - Block diagrams of transmitters and receivers



Passband Data Transmission

- **Coherent Phase-Shift Keying**
 - Here we focus on coherent phase shift keying (PSK) by considering binary PSK, QPSK and its variants, and finish up with M -ary PSK



Passband Data Transmission

- **Binary Phase-Shift Keying - BPSK**

- In a coherent binary PSK system, the pair of signals $s_1(t)$ and $s_2(t)$ used to represent binary symbols 1 and 0, respectively, is defined by:

$$\begin{cases} s_1(t) = \sqrt{\frac{2E_b}{T}} \cos(\omega_c t + \phi) & (\phi = 0) \\ s_2(t) = \sqrt{\frac{2E_b}{T}} \cos(\omega_c t + \phi + \pi) = -s_1(t) \end{cases} \quad 0 \leq t \leq T$$

- where $0 \leq t \leq T_b$ and E_b is the transmitted signal energy per bit



Passband Data Transmission

- **Binary Phase-Shift Keying - BPSK**

- From this pair of equations, it is clear that, in the case of binary PSK, there is only one basis function of unit energy:

$$\psi_1(t) = \sqrt{\frac{2}{T}} \cos \omega_c t$$

- Then we may express the transmitted signals $s_1(t)$ and $s_2(t)$ in terms of $\psi_1(t)$ as follows:

$$s_i(t) = s_{i1}\psi_1(t) = \begin{cases} s_1(t) = s_{11}\psi_1(t) = \sqrt{E_b}\psi_1(t) \\ s_2(t) = s_{21}\psi_1(t) = -\sqrt{E_b}\psi_1(t) \end{cases}$$

Passband Data Transmission

- **Binary Phase-Shift Keying - BPSK**

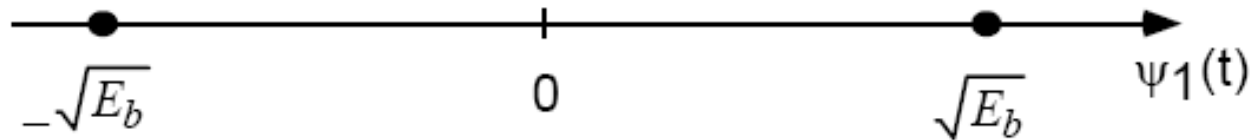
- A coherent binary PSK system is characterized by having a signal space that is one-dimensional ($N = 1$), with a signal constellation consisting of two message points ($M = 2$)
- The coordinates of the message points are:

$$\begin{aligned} s_{11} &= \int_0^{T_b} s_1(t) \phi_1(t) dt \\ &= +\sqrt{E_b} \end{aligned} \qquad \begin{aligned} s_{21} &= \int_0^{T_b} s_2(t) \phi_1(t) dt \\ &= -\sqrt{E_b} \end{aligned}$$

Passband Data Transmission

- **Binary Phase-Shift Keying - BPSK**

- Signal-space diagram for coherent binary PSK system

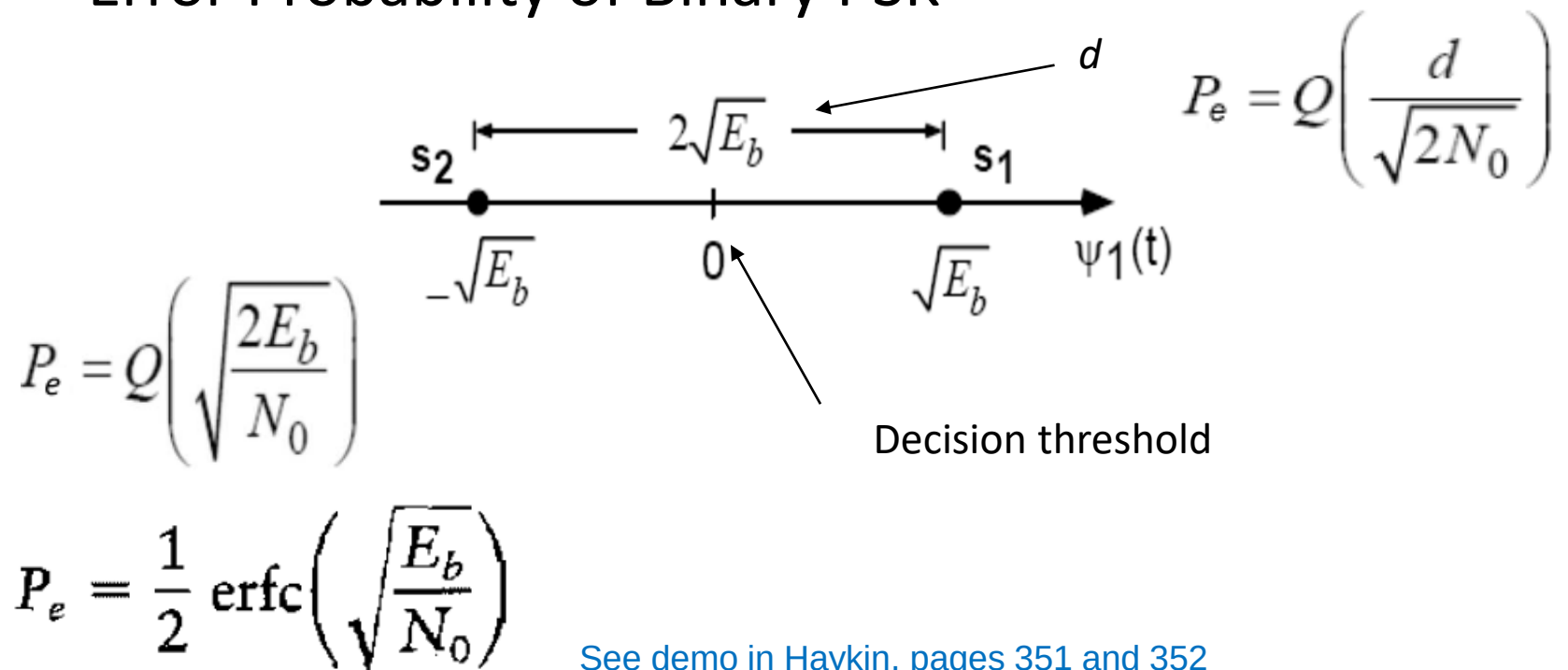


- The message point corresponding to $s_1(t)$ is located at s_{11} and the message point corresponding to $s_2(t)$ is located at s_{21}

Passband Data Transmission

- **Binary Phase-Shift Keying - BPSK**

- Error Probability of Binary PSK



See demo in Haykin, pages 351 and 352

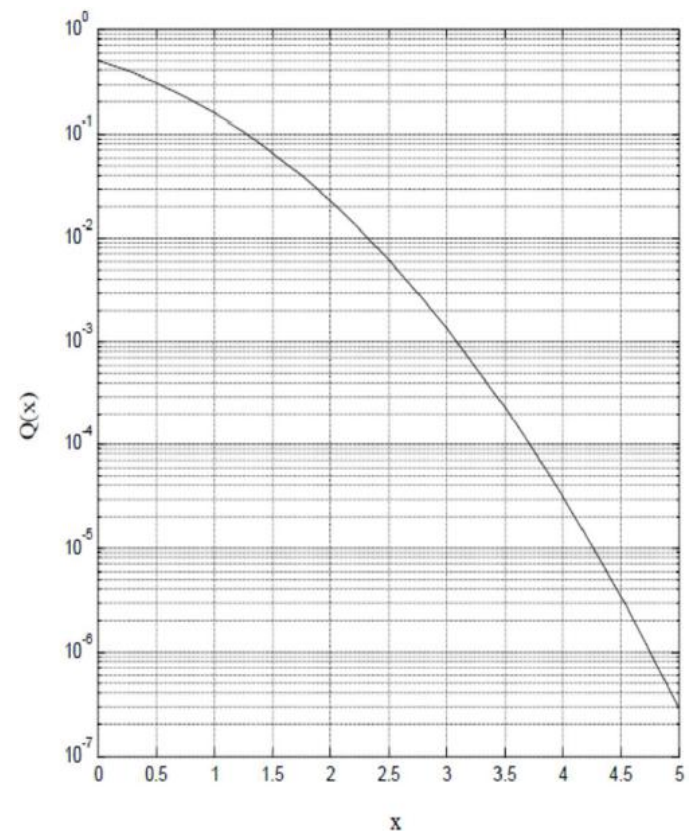
Passband Data Transmission

- ***Q* and *erfc* function**

- Remember:

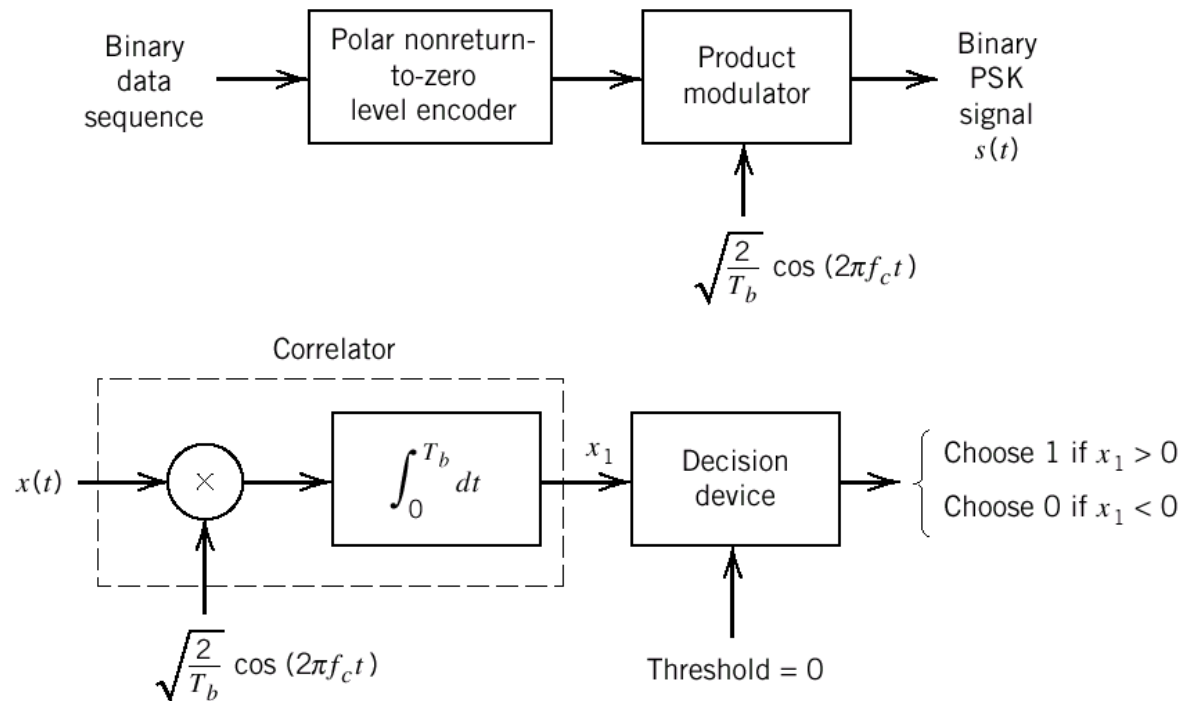
$$Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^{\infty} e^{-\lambda^2/2} d\lambda$$

$$Q(x) = \frac{1}{2} \operatorname{erfc}\left(\frac{x}{\sqrt{2}}\right)$$



Passband Data Transmission

- **Binary Phase-Shift Keying - BPSK**
 - Transmitter and Receiver for Binary PSK



Passband Data Transmission

- **Binary Phase-Shift Keying - BPSK**

- Power Spectra of Binary PSK Signals

- From the modulator of the previous figure, we see that the complex envelope of a binary PSK wave consists of an in-phase component only
 - Furthermore, depending on whether we have symbol 1 or symbol 0 at the modulator input during the signaling interval $0 \leq t \leq T_b$, we find that this in-phase component equals $+g(t)$ or $-g(t)$, respectively



Passband Data Transmission

- **Binary Phase-Shift Keying - BPSK**

- Power Spectra of Binary PSK Signals

- $g(t)$ is the symbol shaping function defined by:

$$g(t) = \begin{cases} \sqrt{\frac{2E_b}{T_b}}, & 0 \leq t \leq T_b \\ 0, & \text{otherwise} \end{cases}$$

- We assume that the input binary wave is random, with symbols 1 and 0 equally likely and the symbols transmitted during the different time slots being statistically independent

Passband Data Transmission

- **Binary Phase-Shift Keying - BPSK**
 - Power Spectra of Binary PSK Signals
 - The power spectral density of a random binary wave is equal to the energy spectral density of the symbol shaping function divided by the symbol duration
 - The energy spectral density of a Fourier transformable signal $g(t)$ is defined as the squared magnitude of the signal's Fourier transform



Passband Data Transmission

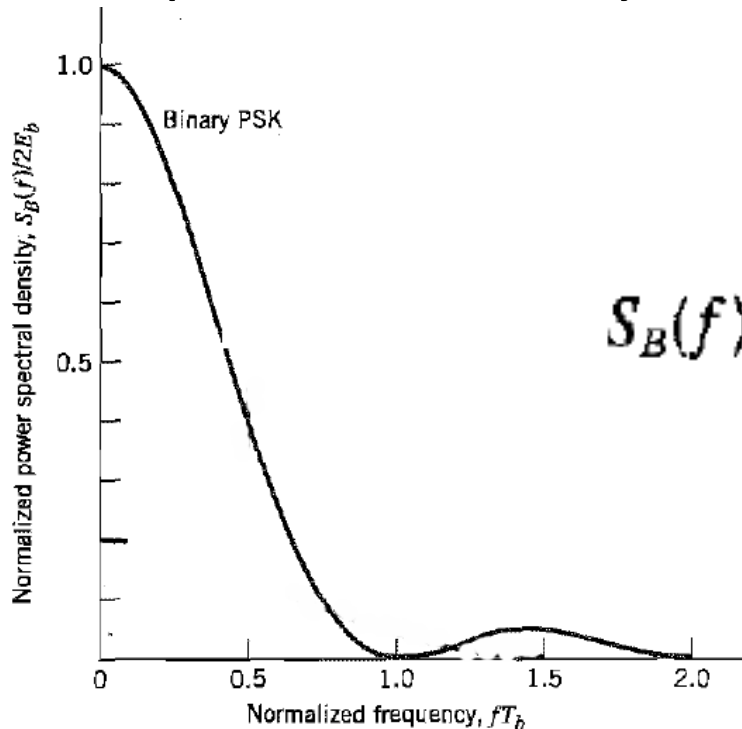
- **Binary Phase-Shift Keying - BPSK**
 - Power Spectra of Binary PSK Signals
 - Hence, the baseband power spectral density of a binary PSK signal equals:

$$\begin{aligned} S_B(f) &= \frac{2E_b \sin^2(\pi T_b f)}{(\pi T_b f)^2} \\ &= 2E_b \operatorname{sinc}^2(T_b f) \end{aligned}$$

Passband Data Transmission

- **Binary Phase-Shift Keying - BPSK**

- Power Spectra of Binary PSK Signals



$$\begin{aligned} S_B(f) &= \frac{2E_b \sin^2(\pi T_b f)}{(\pi T_b f)^2} \\ &= 2E_b \operatorname{sinc}^2(T_b f) \end{aligned}$$

Passband Data Transmission

- **Quadrature-Phase-Shift Keying (QPSK)**

- In QPSK), as with binary PSK, information carried by the transmitted signal is contained in the phase
- In particular, the phase of the carrier takes on one of four equally spaced values: $\pi/4, 3\pi/4, 5\pi/4, 7\pi/4$
- For this set of values, we may define the transmitted signal as:

$$s_i(t) = \sqrt{\frac{2E}{T}} \cos\left(2\pi f_c t - (2i-1)\frac{\pi}{4}\right) \quad \begin{matrix} 0 \leq t \leq T \\ i=1,2,3,4 \end{matrix} \quad f_c = n_c \frac{1}{T}$$

Passband Data Transmission

- **Quadrature Phase Shift Keying (QPSK)**

- There are two orthonormal basis functions, $\psi_1(t)$ and $\psi_2(t)$, contained in the expansion of $s_i(t)$, defined by a pair of quadrature carriers:

$$\text{Basis functions: } \psi_1(t) = \sqrt{\frac{2}{T}} \cos 2\pi f_c t \quad \psi_2(t) = \sqrt{\frac{2}{T}} \sin 2\pi f_c t$$

- There are four message points, and the associated signal vectors are defined by

$$s_i(t) = \underbrace{\sqrt{E} \cos[(2i-1)\pi/4]}_{s_{i1}=s_{Ii}(t)} \psi_1(t) + \underbrace{\sqrt{E} \sin[(2i-1)\pi/4]}_{s_{i2}=s_{Qi}(t)} \psi_2(t)$$



Passband Data Transmission

- **Quadrature-Phase-Shift Keying (QPSK)**

- The elements of the signal vectors, namely, s_{11} and s_{12} , have their values summarized in table:

<i>Gray-encoded Input Dibit</i>	<i>Phase of QPSK Signal (radians)</i>	<i>Coordinates of Message Points</i>	
		s_{11}	s_{12}
10	$\pi/4$	$+\sqrt{E/2}$	$+\sqrt{E/2}$
00	$3\pi/4$	$-\sqrt{E/2}$	$+\sqrt{E/2}$
01	$5\pi/4$	$-\sqrt{E/2}$	$-\sqrt{E/2}$
11	$7\pi/4$	$+\sqrt{E/2}$	$-\sqrt{E/2}$

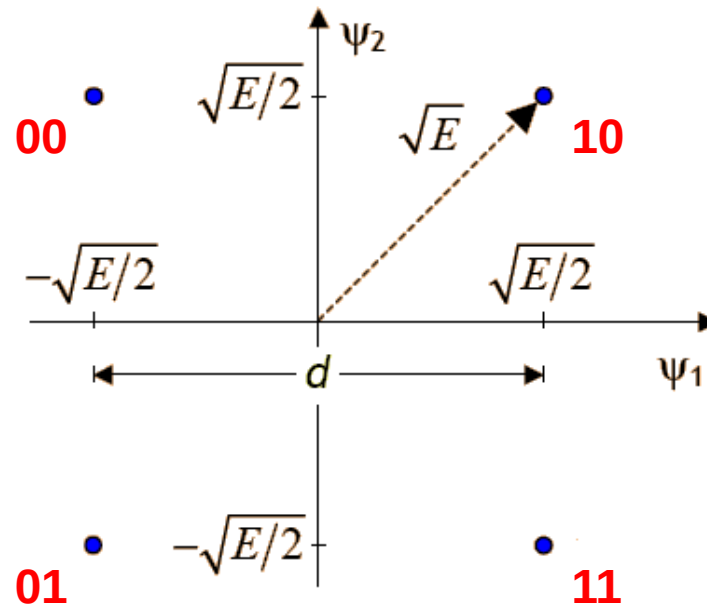
- A QPSK signal has a two-dimensional signal constellation ($N = 2$) and four message points ($M = 4$) whose phase angles increase in a counterclockwise direction



Passband Data Transmission

- **Quadrature Phase-Shift Keying (QPSK)**

- Constellation:



- May be viewed as two orthogonal BPSK signals

Passband Data Transmission

- **Quadrature-Phase-Shift Keying (QPSK)**

- Error Probability of QPSK:
- In a coherent QPSK system, the received signal $x(t)$ is defined by:

$$x(t) = s_i(t) + w(t), \quad \begin{cases} 0 \leq t \leq T \\ i = 1, 2, 3, 4 \end{cases}$$

- where $w(t)$ is the sample function of a white Gaussian noise process of zero mean and power spectral density $N_0/2$



Passband Data Transmission

- **Quadrature-Phase-Shift Keying (QPSK)**
 - Error Probability of QPSK:
 - Correspondingly, the observation vector x has two elements, x_1 and x_2 defined by:

$$\begin{aligned}x_1 &= \int_0^T x(t) \psi_1(t) dt \\&= \sqrt{E} \cos \left[(2i - 1) \frac{\pi}{4} \right] + w_1 \\&= \pm \sqrt{\frac{E}{2}} + w_1\end{aligned}$$

$$\begin{aligned}x_2 &= \int_0^T x(t) \psi_2(t) dt \\&= -\sqrt{E} \sin \left[(2i - 1) \frac{\pi}{4} \right] + w_2 \\&= \mp \sqrt{\frac{E}{2}} + w_2\end{aligned}$$

Passband Data Transmission

- **Quadrature Phase Shift Keying (QPSK)**
 - Error Probability of QPSK:
 - Thus, the observable elements x_1 and x_2 are sample values of independent Gaussian random variables with mean values equal to $\pm\sqrt{E/2}$ and $\mp\sqrt{E/2}$, respectively, and with a common variance equal to $N_0/2$
 - Decision rule: decide that $s_1(t)$ was transmitted if the received signal point associated with the observation vector $\mathbf{x}$ falls inside region Z_1 , decide that $s_2(t)$ was transmitted if the received signal point falls inside region Z_2 , and so on



Passband Data Transmission

- **Quadrature-Phase-Shift Keying (QPSK)**
 - Error Probability of QPSK:
 - To calculate the average probability of symbol error, we note that a coherent QPSK system is equivalent to 2 coherent binary PSK systems working in parallel and using 2 carriers that are in phase quadrature
 - These 2 binary PSK systems may be characterized as follows:
 - The signal energy per bit is $E/2$
 - The noise spectral density is $N_0/2$



Passband Data Transmission

- **Quadrature Phase-Shift Keying (QPSK)**

- Error Probability of QPSK:
- Therefore, a coherent QPSK system achieves the same average probability of bit error as a coherent binary PSK system for the same bit rate and the same E_b/N_0 , but uses only half the channel bandwidth

Symbol probability of error

$$P_e (QPSK) = \frac{P_e}{2} \approx Q\left(\sqrt{\frac{2E_b}{N_0}}\right) = P_e (BPSK)$$

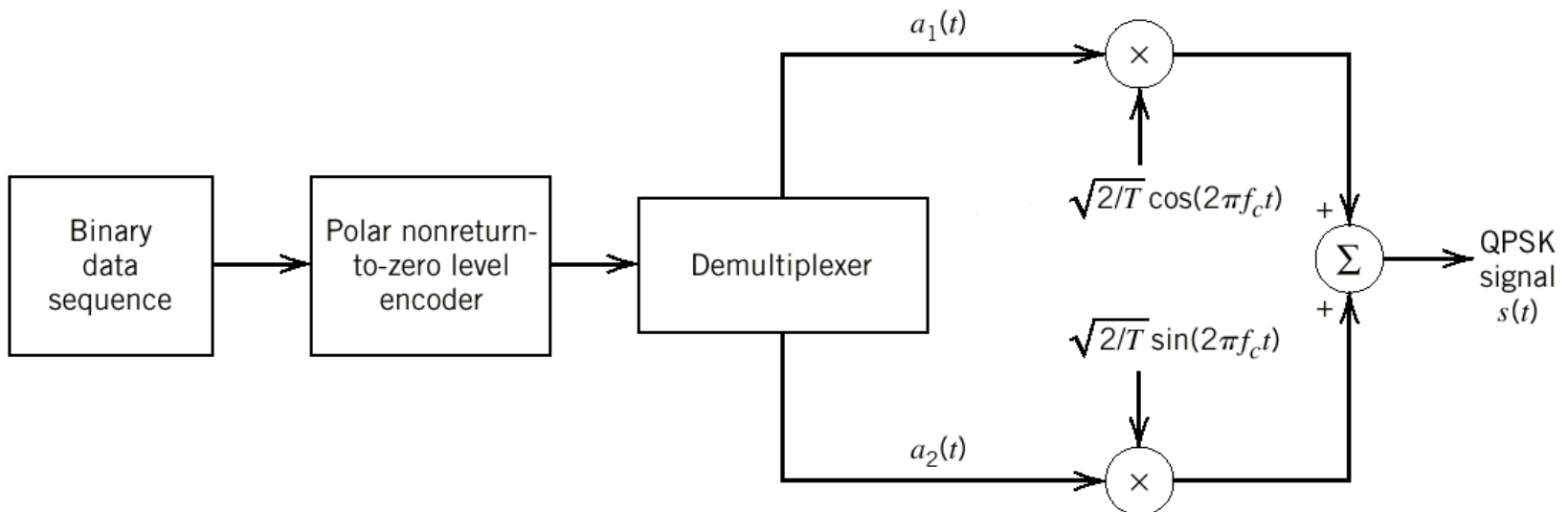
Passband Data Transmission

- **Quadrature-Phase-Shift Keying (QPSK)**
 - Error Probability of QPSK:
 - Stated in a different way, for the same E_b/N_0 and therefore the same average probability of bit error, a coherent QPSK system transmits information at twice the bit rate of a coherent binary PSK system for the same channel bandwidth
 - For a prescribed performance, QPSK uses channel bandwidth better than binary PSK, which explains the preferred use of QPSK over binary PSK in practice



Passband Data Transmission

- **Quadrature Phase-Shift Keying (QPSK)**
 - QPSK Transmitter



Passband Data Transmission

- **Quadrature Phase-Shift Keying (QPSK)**
 - QPSK Transmitter
 - The incoming binary data is first transformed into polar form by a Polar NRZ encoder
 - Symbols 1 and 0 are represented by $+\sqrt{E_b}$ and $-\sqrt{E_b}$ respectively
 - Binary wave is next divided by means of a demultiplexer into 2 separate binary waves consisting of the odd-and even numbered input bits
 - These 2 binary waves are denoted by $a_1(t)$ and $a_2(t)$



Passband Data Transmission

- **Quadrature-Phase-Shift Keying (QPSK)**
 - QPSK Transmitter
 - In any signaling interval, the amplitudes of $a_1(t)$ and $a_2(t)$ equal s_{11} and s_{12} , respectively, depending on the particular dibit that is being transmitted
 - The 2 binary waves $a_1(t)$ and $a_2(t)$ are used to modulate a pair of quadrature carriers or orthonormal basis functions $\psi_1(t)$ and $\psi_2(t)$



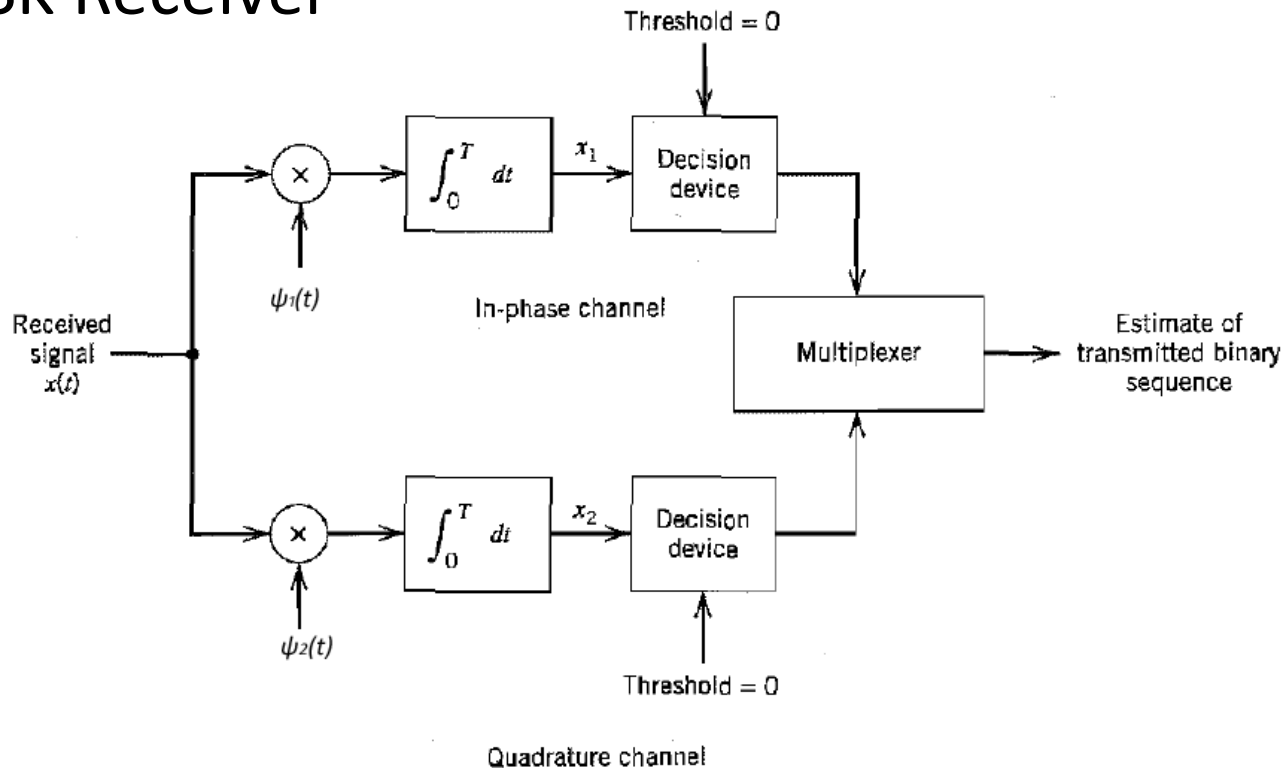
Passband Data Transmission

- **Quadrature-Phase-Shift Keying (QPSK)**
 - QPSK Transmitter
 - The result is a pair of binary PSK signals, which may be detected independently due to the orthogonality of $\psi_1(t)$ and $\psi_2(t)$
 - Finally, the 2 binary PSK signals are added to produce the desired QPSK signal



Passband Data Transmission

- **Quadriphase-Shift Keying (QPSK)**
 - QPSK Receiver



Passband Data Transmission

- **Quadrature-Phase-Shift Keying (QPSK)**
 - QPSK Receiver
 - The QPSK receiver consists of a pair of correlators with a common input and supplied with a locally generated pair of coherent reference signals $\psi_1(t)$ and $\psi_2(t)$
 - Correlator outputs x_1 and x_2 in response to the received signal $x(t)$ and are each compared with a threshold of zero
 - If $x_1 > 0$, a decision is made in favor of symbol 1 for the in-phase channel output
 - if $x_1 < 0$, a decision is made in favor of symbol 0



Passband Data Transmission

- **Quadrature-Phase-Shift Keying (QPSK)**
 - QPSK Receiver
 - Similarly for the quadrature channel output:
 - if $x_2 > 0$, a decision is made in favor of symbol 1, but if $x_2 < 0$, a decision is made in favor of symbol 0
 - Finally, these 2 binary sequences, at the in-phase and quadrature channel outputs, are combined in a multiplexer to reproduce the original binary sequence at the transmitter input with the minimum probability of symbol error in an AWGN channel



Passband Data Transmission

- **Quadrature-Phase-Shift Keying (QPSK)**
 - Power Spectra of QPSK Signals
 - We make the following observations pertaining to the in-phase and quadrature components of a QPSK signal:
 1. Depending on the dibit sent during the signaling interval $-T_b \leq t \leq T_b$, the in-phase component equals $+g(t)$ or $-g(t)$, and similarly for the quadrature component



Passband Data Transmission

- **Quadrature Phase-Shift Keying (QPSK)**

- Power Spectra of QPSK Signals
- The $g(t)$ denotes the symbol shaping function, defined by:

$$g(t) = \begin{cases} \sqrt{\frac{E}{T}}, & 0 \leq t \leq T \\ 0, & \text{otherwise} \end{cases}$$

- Hence, the in-phase and quadrature components have a common power spectral density, namely: $E \cdot \text{sinc}^2(Tf)$

Passband Data Transmission

- **Quadrature-Phase-Shift Keying (QPSK)**

- Power Spectra of QPSK Signals

2. The in-phase and quadrature components are statistically independent

- Accordingly, the power spectral density of the QPSK signal equals the sum of the individual power spectral densities of the in-phase and quadrature components, so we may write:

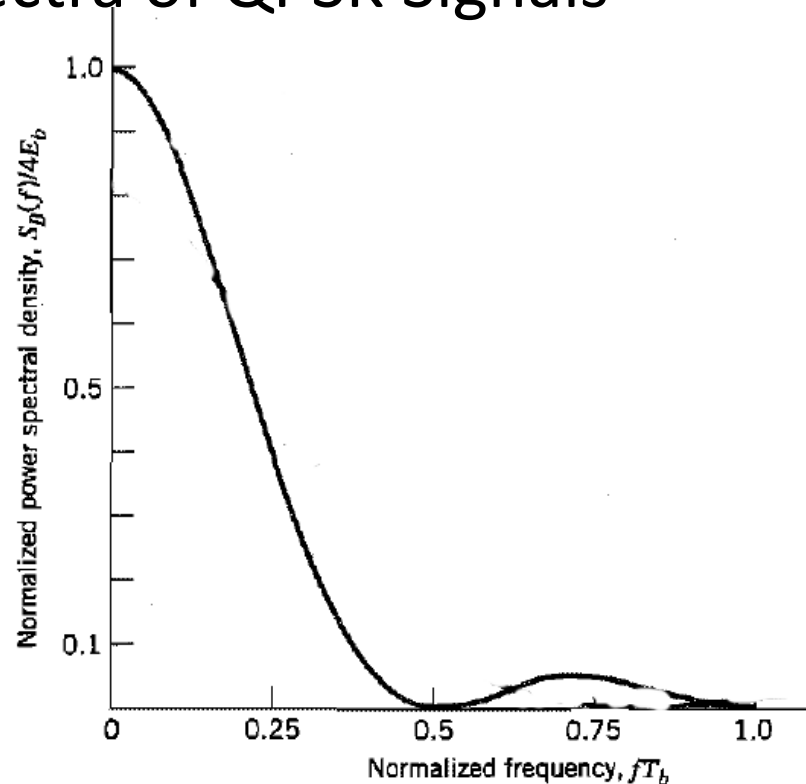
$$S_B(f) = 2E_s \text{sinc}^2(Tf) = 4E_b \text{sinc}^2(2T_b f)$$

- For the same bit rate QPSK uses half bandwidth of BPSK



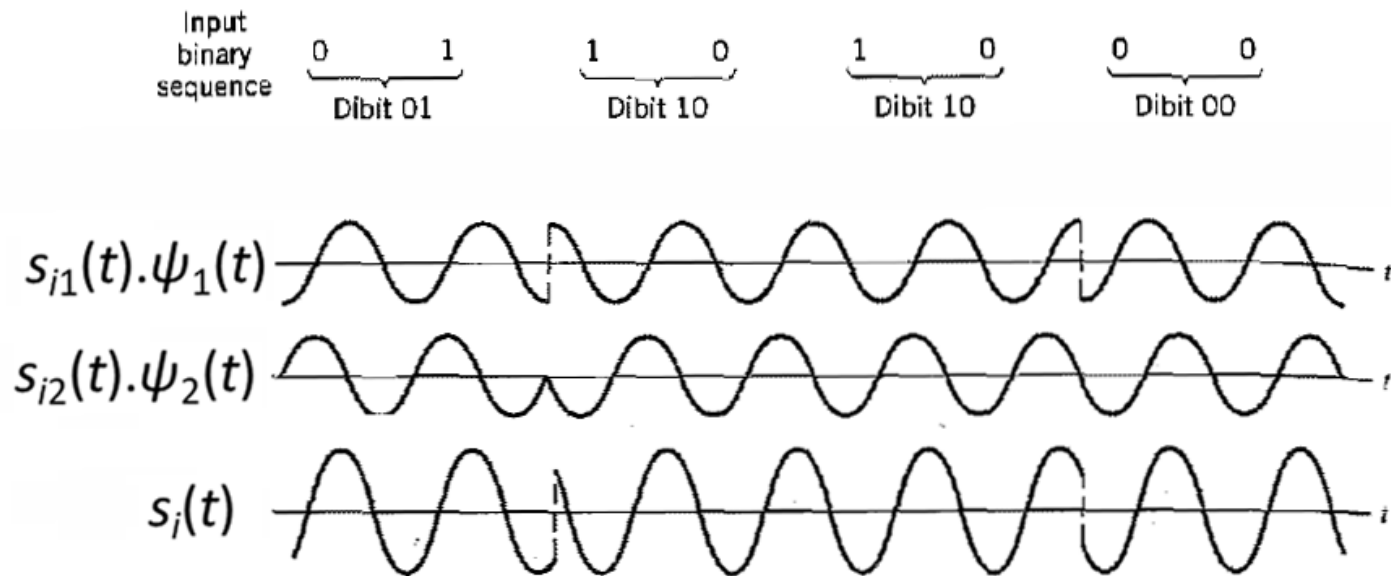
Passband Data Transmission

- **Quadrature Phase-Shift Keying (QPSK)**
 - Power Spectra of QPSK Signals



Passband Data Transmission

- **Quadrature Phase-Shift Keying (QPSK)**
 - Offset QPSK
 - Examining the QPSK waveforms:



Passband Data Transmission

- **Quadrature-Shift Keying (QPSK)**
 - Offset QPSK
 1. The carrier phase changes by ± 180 degrees whenever both the in-phase and quadrature components of the QPSK signal changes sign
 - when the input binary sequence switches from dibit 01 to dibit 10
 2. The carrier phase changes by ± 90 degrees whenever the in-phase or quadrature component changes sign
 - when the input binary sequence switches from dibit 10 to dibit 00, during which the in-phase component changes sign, whereas the quadrature component is unchanged



Passband Data Transmission

- **Quadriphase-Shift Keying (QPSK)**

- Offset QPSK

- 3. The carrier phase is unchanged when neither the in-phase component nor the quadrature component changes sign
 - when dibit 10 is transmitted in two successive symbol intervals

- Situation 1 and 2 can be of a particular concern when the QPSK signal is filtered during transmission

- 180 and 90 degree shifts in carrier phase can result in changes in the carrier amplitude (i.e., envelope of the QPSK signal), causing additional symbol errors on detection



Passband Data Transmission

- **Quadrature Phase Shift Keying (QPSK)**

- Offset QPSK

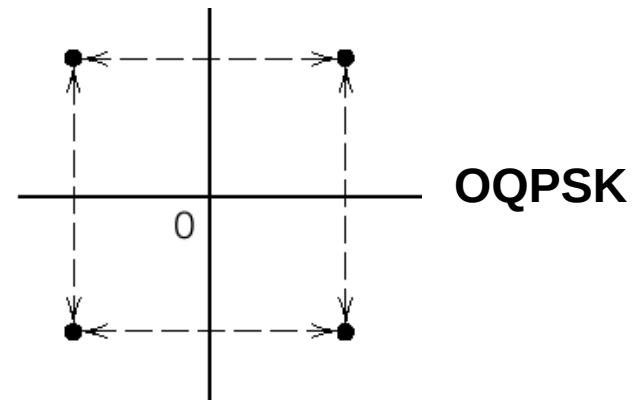
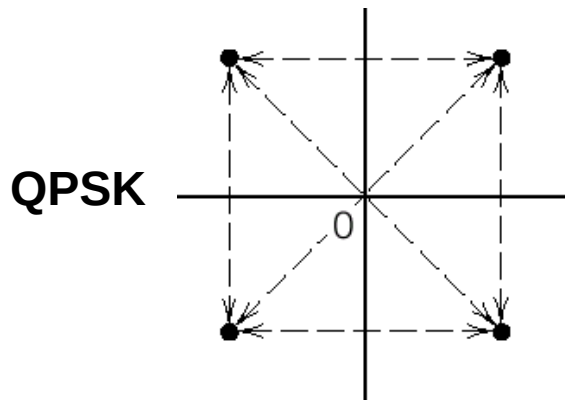
- In this variant of QPSK, the bit stream responsible for generating the quadrature component is delayed (i.e., offset) by half a symbol interval with respect to the bit stream responsible for generating the in-phase component

$$\begin{aligned}\psi_1(t) &= \sqrt{\frac{2}{T}} \cos(2\pi f_c t), & 0 \leq t \leq T \\ \psi_2(t) &= \sqrt{\frac{2}{T}} \sin(2\pi f_c t), & \frac{T}{2} \leq t \leq \frac{3T}{2}\end{aligned}$$



Passband Data Transmission

- **Quadrature Phase-Shift Keying (QPSK)**
 - Offset QPSK
 - Doing this, the possible paths for switching between the message points are:



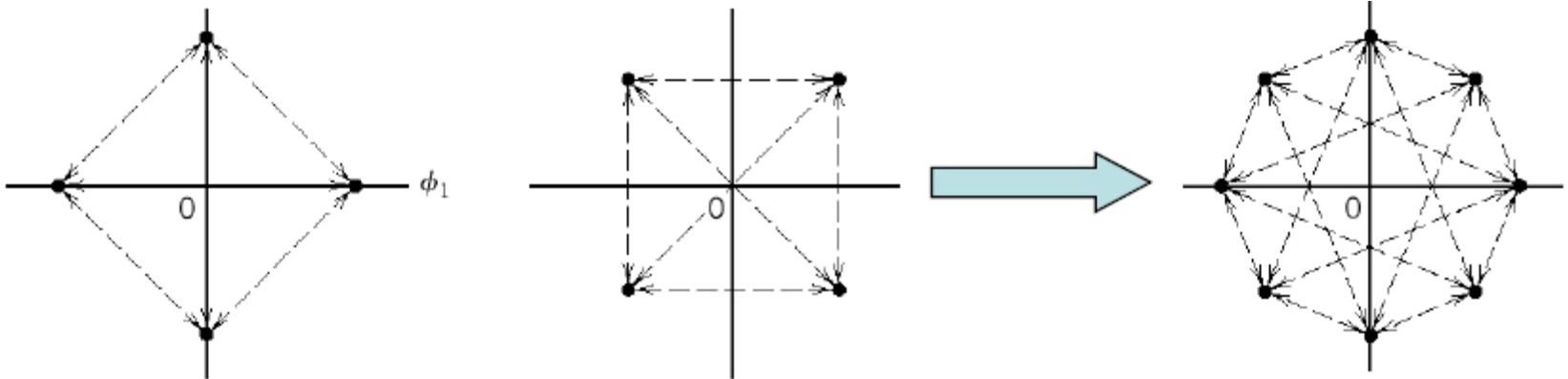
- Phase transitions are confined to $\pm 90^\circ$ which implies less amplitude fluctuations after filtering

Passband Data Transmission

- **Quadrature-Phase-Shift Keying (QPSK)**

- $\pi/4$ Shifted QPSK

- Phase of successive symbols is alternatively picked from the two constellations below:



Passband Data Transmission

- **Quadrature-Phase-Shift Keying (QPSK)**
 - $\pi/4$ Shifted QPSK
 - There are eight possible states in the constellation
 - Phase transition in successive symbols are restricted to $\pm\pi/4$ and $\pm3\pi/4$
 - Same probability of bit error as QPSK
 - Consequently, envelope variations of $\pi/4$ -shifted QPSK signals due to filtering are significantly reduced, compared to those in QPSK



Passband Data Transmission

- ***M*-ary PSK (MPSK)**

- QPSK is a special case of *M*-ary PSK, where the phase of the carrier takes on one of *M* possible values, namely, $\varphi_i = 2(i - 1)\pi/M$, where $i = 1, 2, \dots, M$
- Accordingly, during each signaling interval of duration *T*, one of the *M* possible signals:

$$s_i(t) = \sqrt{\frac{2E}{T}} \cos\left(2\pi f_c t + \frac{2\pi}{M} (i - 1)\right), \quad i = 1, 2, \dots, M$$

- is sent

Passband Data Transmission

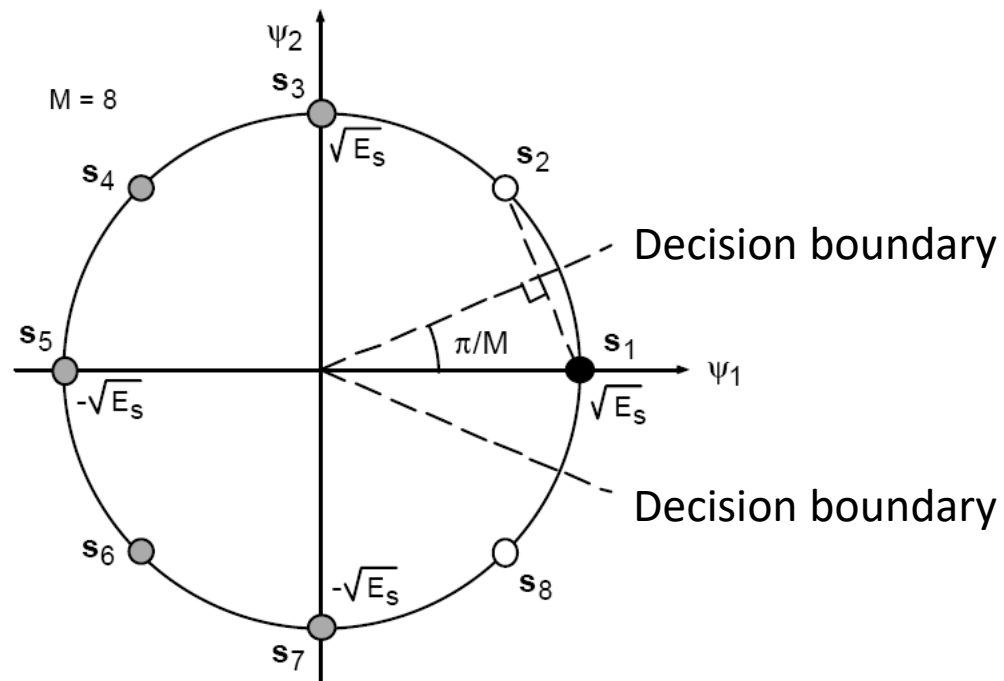
- ***M*-ary PSK (MPSK)**
 - Each $S_i(t)$ may be expanded in terms of the same two basis functions $\psi_1(t)$ and $\psi_2(t)$
 - The signal constellation of *M*-ary PSK is therefore two-dimensional
 - The *M* message points are equally spaced on a circle of radius $\sqrt{E}$ and center at the origin



Passband Data Transmission

- ***M*-ary PSK (MPSK)**

- Constellation for the case of octaphaseshift-keying (i.e., $M = 8$)



Passband Data Transmission

- ***M*-ary PSK (MPSK)**

- The Euclidean distance of each of these two points from m_1 is (for $M = 8$):

$$d_{12} = d_{18} = 2\sqrt{E_s} \sin\left(\frac{\pi}{M}\right)$$

- Hence, the use of Equations of slide 22 yields the average probability of symbol error for coherent M -ary PSK as:

$$P_e \approx Q\left(\frac{d_{12}}{\sqrt{2N_0}}\right) + Q\left(\frac{d_{18}}{\sqrt{2N_0}}\right) = 2Q\left(\sqrt{\frac{2E_s}{N_0}} \sin\frac{\pi}{M}\right)$$

Passband Data Transmission

- ***M*-ary PSK (MPSK)**

- For $M = 4$, the equation reduces to the same form given in equation of slide 38 for QPSK

$$P_e \approx Q\left(\frac{d_{12}}{\sqrt{2N_0}}\right) + Q\left(\frac{d_{18}}{\sqrt{2N_0}}\right) = 2Q\left(\sqrt{\frac{2E_s}{N_0}} \sin \frac{\pi}{M}\right)$$

$$P_b(QPSK) = \frac{P_e}{2} \approx Q\left(\sqrt{\frac{2E_b}{N_0}}\right) = P_b(BPSK)$$

Passband Data Transmission

- ***M*-ary PSK (MPSK)**

- The symbol duration of *M*-ary PSK is defined by:

$$T = T_b \log_2 M$$

- where T_b is the bit duration

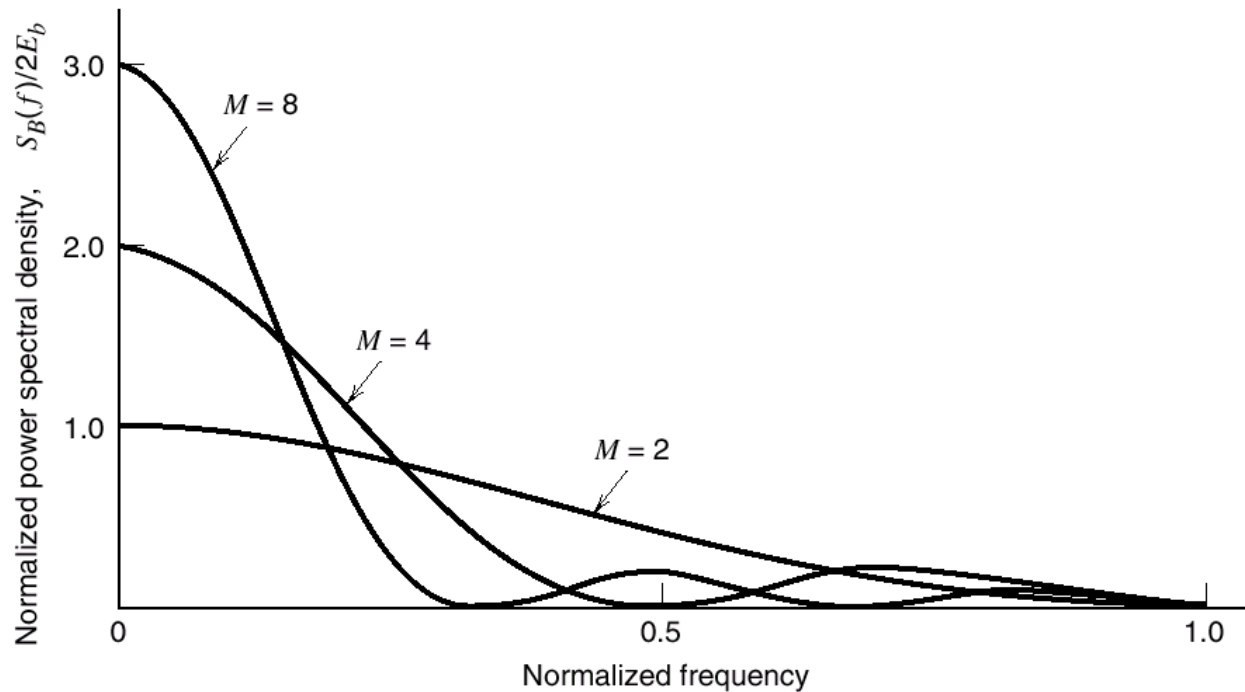
- Proceeding in a manner similar to that described for a QPSK signal, we may show that the baseband power spectral density of an *M*-ary PSK signal is given by:

$$\begin{aligned} S_B(f) &= 2E_s \operatorname{sinc}^2(Tf) = \\ &= 2E_b \log_2 M \operatorname{sinc}^2(T_b f \log_2 M) \end{aligned}$$



Passband Data Transmission

- ***M*-ary PSK (MPSK)**
 - MPSK Power Spectra



Passband Data Transmission

- ***M*-ary PSK (MPSK)**

- Bandwidth efficiency for MPSK
- The channel bandwidth required to pass *M*-ary PSK signals (i.e. the main spectral lobe of *M*-ary signals) is given by:

$$B = \frac{2}{T}$$

- where *T* is the symbol duration
- But the symbol duration *T* is related to the bit duration T_b by:

$$T = T_b \log_2 M$$



Passband Data Transmission

- ***M*-ary PSK (MPSK)**

- Bandwidth efficiency for MPSK
- Moreover, the bit rate $R_b = 1/T_b$
- Hence, we may redefine the channel bandwidth in terms of the bit rate R_b as:

$$B = \frac{2R_b}{\log_2 M}$$

- Based on this, the bandwidth efficiency of *M*-ary PSK signals is given by:

$$\rho = \frac{R_b}{B} = \frac{\log_2 M}{2}$$



Passband Data Transmission

- ***M*-ary PSK (MPSK)**

- Bandwidth efficiency for MPSK

M	2	4	8	16	32	64
ρ (bits/s/Hz)	0.5	1	1.5	2	2.5	3

- As the number of states M is increased, ρ is improved at the expense of error performance
 - To ensure that there is no degradation in error performance, we have to increase E_b/N_0 to compensate for the increase in M

Passband Data Transmission

- **Hybrid Amplitude/Phase Modulation Schemes**
 - In an M -ary PSK system, the in-phase and quadrature components of the modulated signal are interrelated in such a way that the envelope is constrained to remain constant
 - This constraint manifests itself in a circular constellation for the message points
 - However, if this constraint is removed, and the in-phase and quadrature components are thereby permitted to be independent, we get a new modulation scheme called M -ary quadrature amplitude modulation (QAM)
 - This latter modulation scheme is hybrid in nature in that the carrier experiences amplitude as well as phase modulation



Passband Data Transmission

- **Hybrid Amplitude/Phase Modulation Schemes**
 - M -ary Quadrature Amplitude Modulation (QAM)
 - In T2 session, we studied M -ary PAM which is one-dimensional
 - M -ary QAM is a 2-dimensional generalization of M -ary PAM in that its formulation involves 2 orthogonal passband basis functions:
$$\begin{cases} \psi_1(t) = \sqrt{\frac{2}{T}} \cos 2\pi f_c t \\ \psi_2(t) = \sqrt{\frac{2}{T}} \sin 2\pi f_c t \end{cases} \quad 0 \leq t \leq T$$



Passband Data Transmission

- **Hybrid Amplitude/Phase Modulation Schemes**
 - M -ary Quadrature Amplitude Modulation (QAM)
 - Let the i th message point s_i in the $(\psi_1$ and $\psi_2)$ plane be denoted by $(a_i \cdot d_{min}/2, b_i \cdot d_{min}/2)$
 - where d_{min} is the minimum distance between any two message points in the constellation, a , and b , are integers, and: $i = 1, 2, \dots, M$
 - Let $d_{min}/2 = \sqrt{E_0}$, where E_0 is the energy of the signal with the lowest amplitude



Passband Data Transmission

- **Hybrid Amplitude/Phase Modulation Schemes**
 - *M*-ary Quadrature Amplitude Modulation (QAM)
 - The transmitted *M*-ary QAM signal for symbol *i*, is then defined by:

$$s_i(t) = \sqrt{\frac{2E_0}{T}} a_i \cos 2\pi f_c t + \sqrt{\frac{2E_0}{T}} b_i \sin 2\pi f_c t \quad \begin{matrix} 0 \leq t \leq T \\ i = -L+1, \dots, -1, 0, 1, \dots, L-1 \end{matrix}$$

- The signal $s_i(t)$ consists of two phase-quadrature carriers with each one being modulated by a set of discrete amplitudes, hence the name quadrature amplitude modulation - QAM



Passband Data Transmission

- **Hybrid Amplitude/Phase Modulation Schemes**
 - *M*-ary Quadrature Amplitude Modulation (QAM)

$$s_i(t) = \sqrt{\frac{2E_0}{T}} a_i \cos 2\pi f_c t + \sqrt{\frac{2E_0}{T}} b_i \sin 2\pi f_c t \quad \begin{matrix} 0 \leq t \leq T \\ i = -L + 1, \dots, -1, 0, 1, \dots, L - 1 \end{matrix}$$

$$\begin{cases} \psi_1(t) = \sqrt{\frac{2}{T}} \cos 2\pi f_c t \\ \psi_2(t) = \sqrt{\frac{2}{T}} \sin 2\pi f_c t \end{cases} \quad 0 \leq t \leq T$$

$$s_i(t) = a_i \sqrt{E_0} \psi_1(t) + b_i \sqrt{E_0} \psi_2(t) \quad \begin{matrix} 0 \leq t \leq T \\ i = -L + 1, \dots, -1, 0, 1, \dots, L - 1 \end{matrix}$$



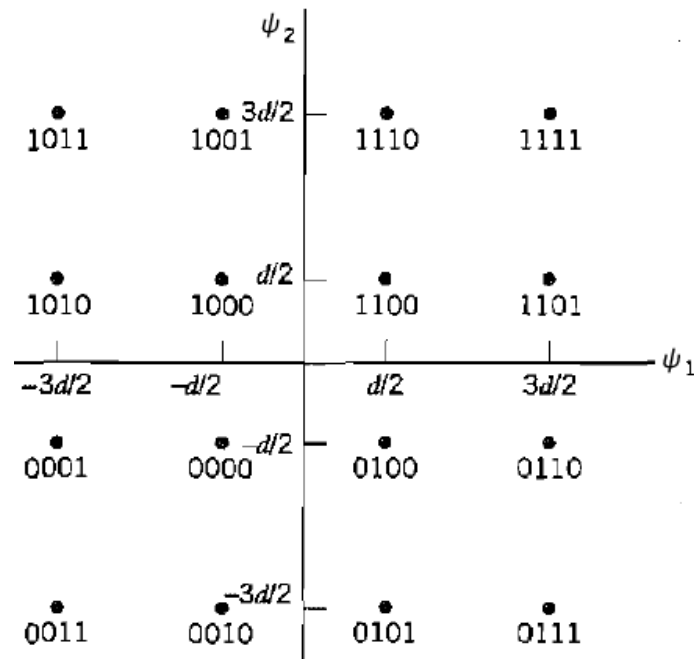
Passband Data Transmission

- **Hybrid Amplitude/Phase Modulation Schemes**
 - M -ary Quadrature Amplitude Modulation (QAM)
 - Depending on the number of possible symbols M , we may distinguish two distinct QAM constellations:
 - square constellations for which the number of bits per symbol is even
 - cross constellations for which the number of bits per symbol is odd



Passband Data Transmission

- **Hybrid Amplitude/Phase Modulation Schemes**
 - Example of a QAM Square Constellations – $L = \sqrt{M}$
 - $M = 4$ – 16QAM with gray encoding



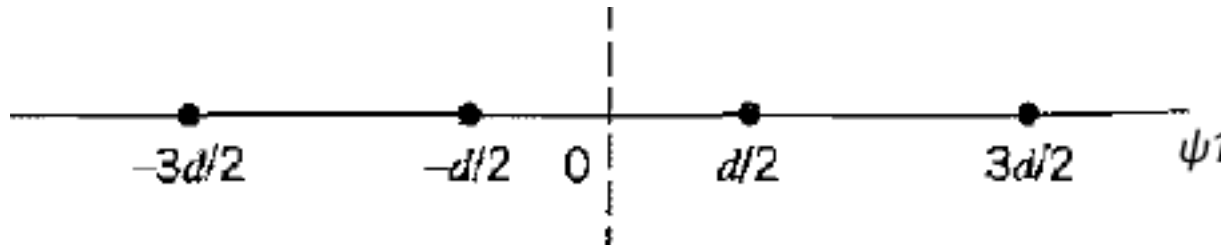
Passband Data Transmission

- **Hybrid Amplitude/Phase Modulation Schemes**
 - QAM Square Constellations
 - 2 of the 4 bits, namely, the left-most 2 bits, specify the quadrant in the $(\psi_1$ and $\psi_2)$ plane in which a message point lies
 - Starting from the first quadrant and proceeding counterclockwise, the 4 quadrants are represented by the dibits 11, 10, 00, and 01
 - The remaining 2 bits are used to represent 1 of the 4 possible symbols lying within each quadrant of the $(\psi_1$ and $\psi_2)$ plane
 - The encoding of the 4 quadrants and also the encoding of the symbols in each quadrant follow the Gray coding rule



Passband Data Transmission

- **Hybrid Amplitude/Phase Modulation Schemes**
 - QAM Square Constellations
 - For the example at hand, we have $L = 4$
 - Thus, the square constellation is the Cartesian product of the 4-PAM constellation with itself



Passband Data Transmission

- **Hybrid Amplitude/Phase Modulation Schemes**
 - QAM Square Constellations
 - Moreover, the matrix of the constellation has the value:

$$\{a_i, b_i\} = \begin{bmatrix} (-3, 3) & (-1, 3) & (1, 3) & (3, 3) \\ (-3, 1) & (-1, 1) & (1, 1) & (3, 1) \\ (-3, -1) & (-1, -1) & (1, -1) & (3, -1) \\ (-3, -3) & (-1, -3) & (1, -3) & (3, -3) \end{bmatrix}$$

Passband Data Transmission

- **Hybrid Amplitude/Phase Modulation Schemes**
 - QAM Square Constellations
 - To calculate the probability of symbol error for M -ary QAM, we exploit the property that a QAM square constellation can be factored into the product of the corresponding PAM constellation with itself
 - We may thus proceed as follows:
 - 1. The probability of correct detection for M -ary QAM may be written as: $P_e = (1 - P_e')^2$



Passband Data Transmission

- **Hybrid Amplitude/Phase Modulation Schemes**

- QAM Square Constellations

- where P_e' is the probability of symbol error for the corresponding L -ary PAM with $L = \sqrt{M}$

- 2. The probability of symbol error P_e' is defined by:

$$P_e' = \left(1 - \frac{1}{\sqrt{M}}\right) \operatorname{erfc}\left(\sqrt{\frac{E_0}{N_0}}\right)$$

- 3. The probability of symbol error for M -ary QAM is given by:

$$P_e = 1 - P_c = 1 - (1 - P_e')^2 \approx 2P_e'$$



Passband Data Transmission

- **Hybrid Amplitude/Phase Modulation Schemes**

- QAM Square Constellations

- Putting all together, we find that the probability of symbol error for M -ary QAM is approximately given by:

$$P_e \simeq 2 \left(1 - \frac{1}{\sqrt{M}} \right) \text{erfc} \left(\sqrt{\frac{E_0}{N_0}} \right)$$

- The transmitted energy in M -ary QAM is variable in that its instantaneous value depends on the particular symbol transmitted

Passband Data Transmission

- **Hybrid Amplitude/Phase Modulation Schemes**

- QAM Square Constellations

- It is therefore more logical to express P_e in terms of the average value of the transmitted energy rather than E_0

$$E_{av} = \frac{2(L^2 - 1)E_0}{3} = \frac{2(M - 1)E_0}{3}$$

- And thus:

$$P_e \simeq 2 \left(1 - \frac{1}{\sqrt{M}} \right) \operatorname{erfc} \left(\sqrt{\frac{3E_{av}}{2(M - 1)N_0}} \right)$$

Passband Data Transmission

- **Hybrid Amplitude/Phase Modulation Schemes**
 - QAM Square Constellations

$$E_{\text{av}} = \frac{2(L^2 - 1)E_0}{3} = \frac{2(M - 1)E_0}{3}$$

$$P_e \simeq 2 \left(1 - \frac{1}{\sqrt{M}} \right) \text{erfc} \left(\sqrt{\frac{3E_{\text{av}}}{2(M - 1)N_0}} \right)$$

$$P_e \approx 4 \left(1 - \frac{1}{\sqrt{M}} \right) Q \left(\sqrt{\frac{2E_0}{N_0}} \right) \quad P_e \approx 4 \left(1 - \frac{1}{\sqrt{M}} \right) Q \left(\frac{d_{\min}}{\sqrt{2N_0}} \right)$$

$$P_e \approx 4 \left(1 - \frac{1}{\sqrt{M}} \right) Q \left(\sqrt{\frac{3}{(M - 1)} \frac{\langle E \rangle}{N_0}} \right)$$

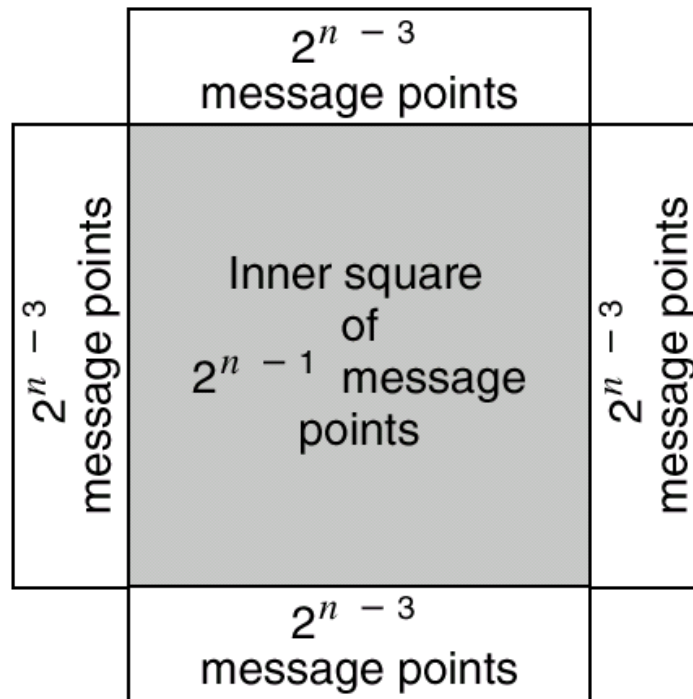
Passband Data Transmission

- **Hybrid Amplitude/Phase Modulation Schemes**
 - QAM Cross Constellations
 - To generate an M -ary QAM signal with an odd number of bits per symbol, we require the use of a cross constellation
 - We may construct such a signal constellation with n bits per symbol by proceeding as follows:
 - Start with a QAM square constellation with $n - 1$ bits per symbol
 - Extend each side of the QAM square constellation by adding 2^{n-3} symbols
 - Ignore the corners in the extension



Passband Data Transmission

- **Hybrid Amplitude/Phase Modulation Schemes**
 - QAM Cross Constellations



2^n symbols in total
(n bit per symbol)

Passband Data Transmission

- **Hybrid Amplitude/Phase Modulation Schemes**
 - QAM Cross Constellations
 - Unlike QAM square constellation, it is not possible to express a QAM cross constellation as the product of a PAM constellation with itself. The absence of such a factorization complicates the determination of the probability of symbol error P_e
 - We therefore simply state the formula for P_e without proof:

$$P_e \approx 2 \left(1 - \frac{1}{\sqrt{2M}} \right) \operatorname{erfc} \left(\sqrt{\frac{E_0}{N_0}} \right) \text{ for high } E_0/N_0$$

Passband Data Transmission

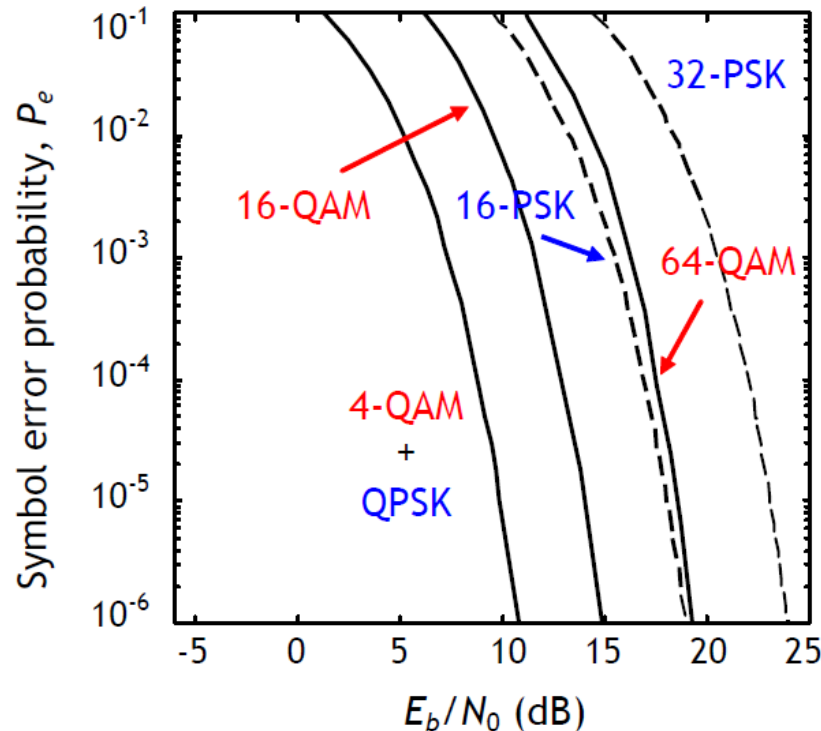
- **Hybrid Amplitude/Phase Modulation Schemes**
 - QAM Cross Constellations
 - Impossible to perfectly gray code this constellation

$$P_e \approx 2 \left(1 - \frac{1}{\sqrt{2M}} \right) \operatorname{erfc} \left(\sqrt{\frac{E_0}{N_0}} \right) \text{ for high } E_0/N_0$$

$$P_e \approx 4 \left(1 - \frac{1}{\sqrt{2M}} \right) Q \left(\sqrt{\frac{2E_0}{N_0}} \right) \text{ for high } E_0/N_0$$

Passband Data Transmission

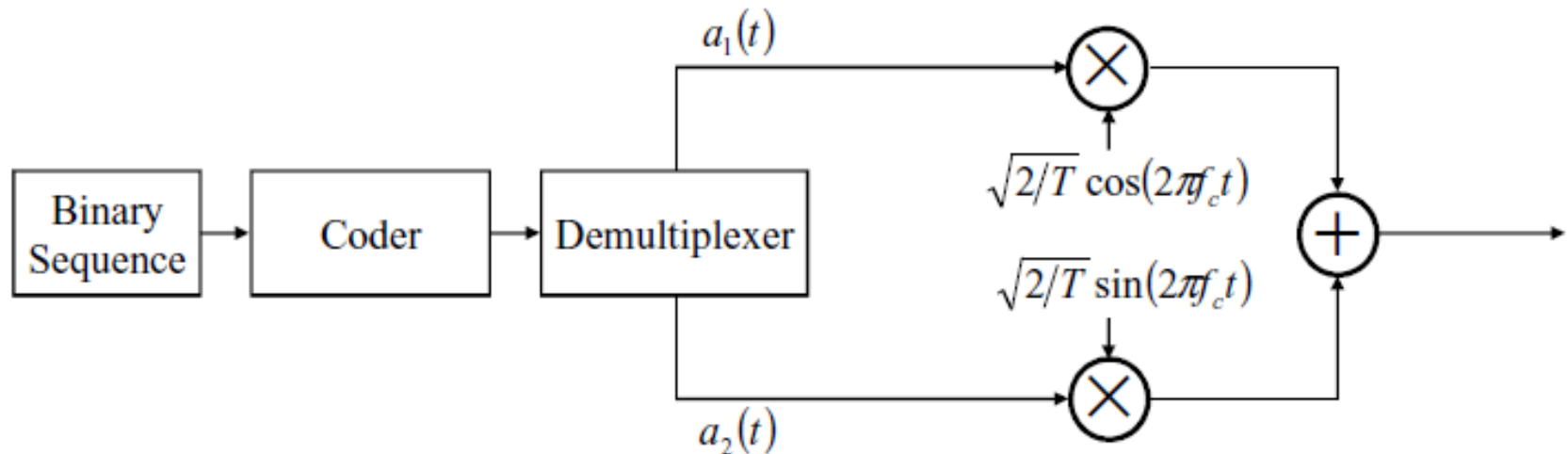
- **Hybrid Amplitude/Phase Modulation Schemes**
 - P_e in MQAM and MPSK



**MQAM
outperforms
MPSK**

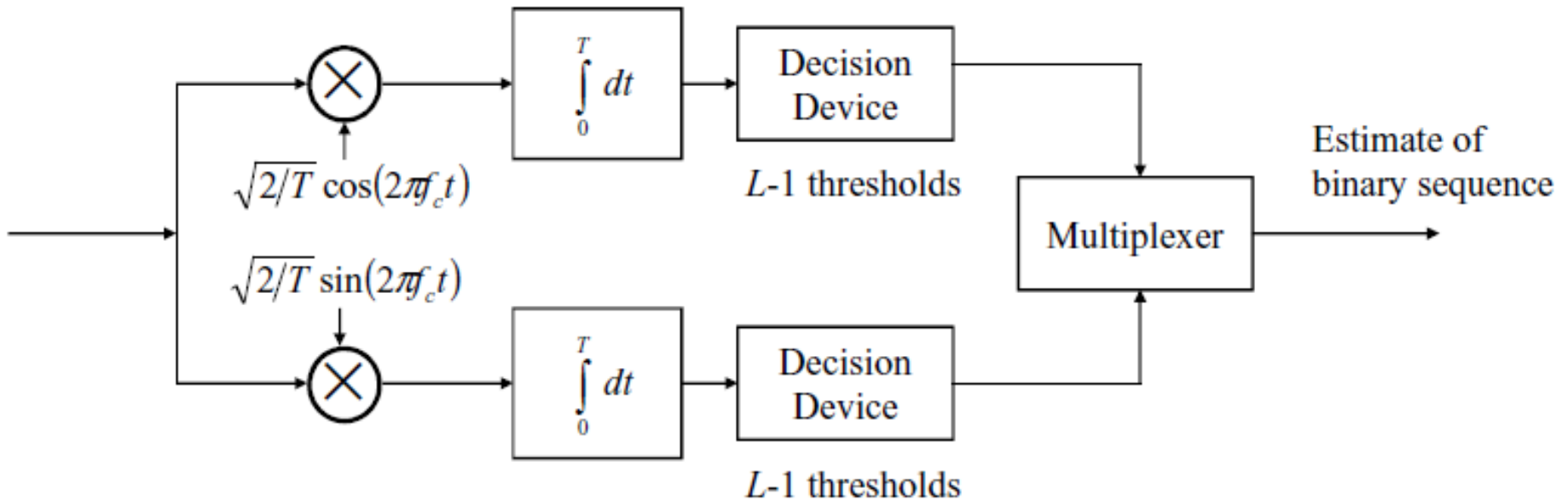
Passband Data Transmission

- **Hybrid Amplitude/Phase Modulation Schemes**
 - QAM Transmitter



Passband Data Transmission

- **Hybrid Amplitude/Phase Modulation Schemes**
 - QAM Receiver



Reading Material

- *Simon Haykin, Communication Systems, McMaster Univ, 5th Edition, 2009 (ISBN 978-047-16979-0-9)*
- *A. Bruce Carlson and Paul Crilly. Communication Systems – An Introduction to Signals and Noise, McGraw-Hill, 5th edition, 2009 (ISBN 978-007-338-040-7)*
- *Bernard Sklar and Fredric Harris, Digital Communications: Fundamentals and Applications, Pearson, 3rd Edition, 2020 (ISBN-13: 978-0134588568)*



Reading Material

